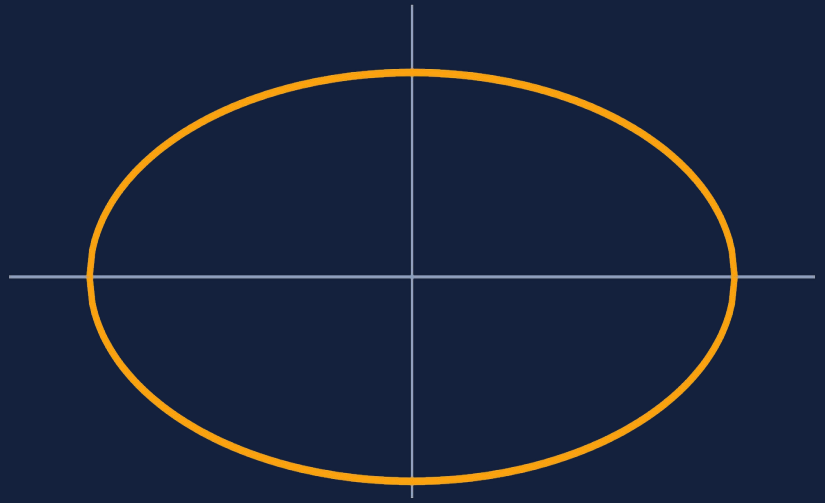
Root Locus Technique



Control Systems (22EEG53)

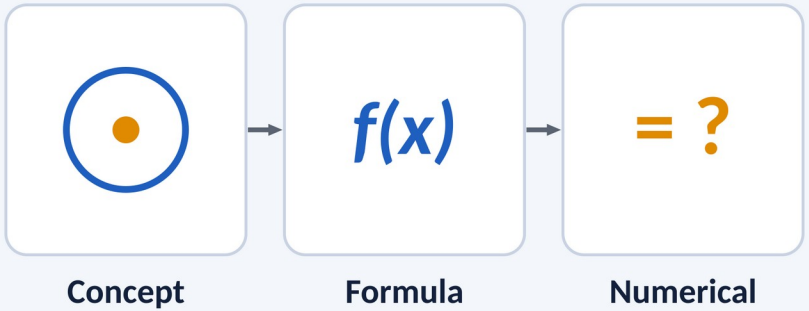
Unit 4: Stability of Systems

Topic: Root locus concept and rules for construction

Prof. Mitesh Astik

Learning Outcomes

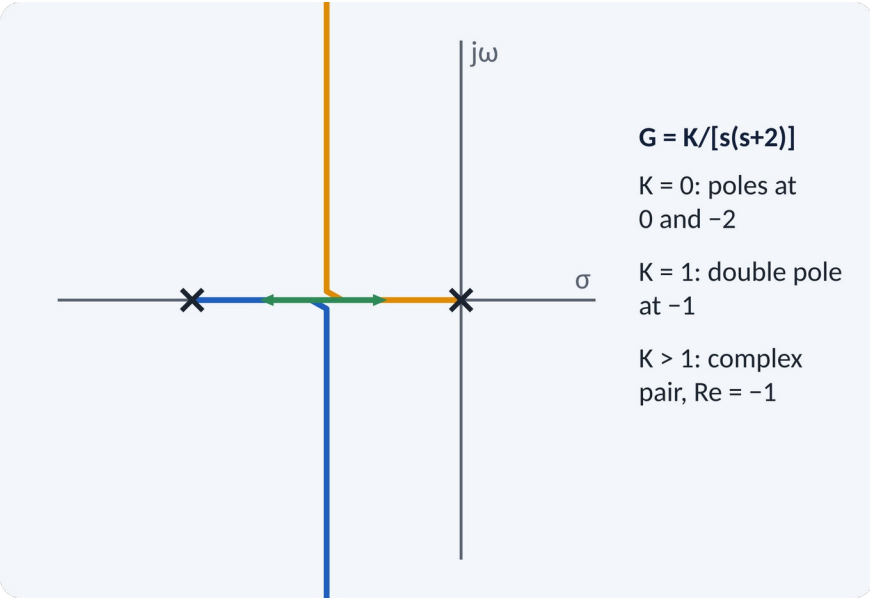
- State the root-locus idea and its conditions.
- Apply the construction rules step by step.
- Find breakaway points and the jω-axis crossing.
- Pick the gain K for a desired damping ratio.



Linked to CO4
Understand stability using Routh–Hurwitz and root locus.

Recap: What You Already Know

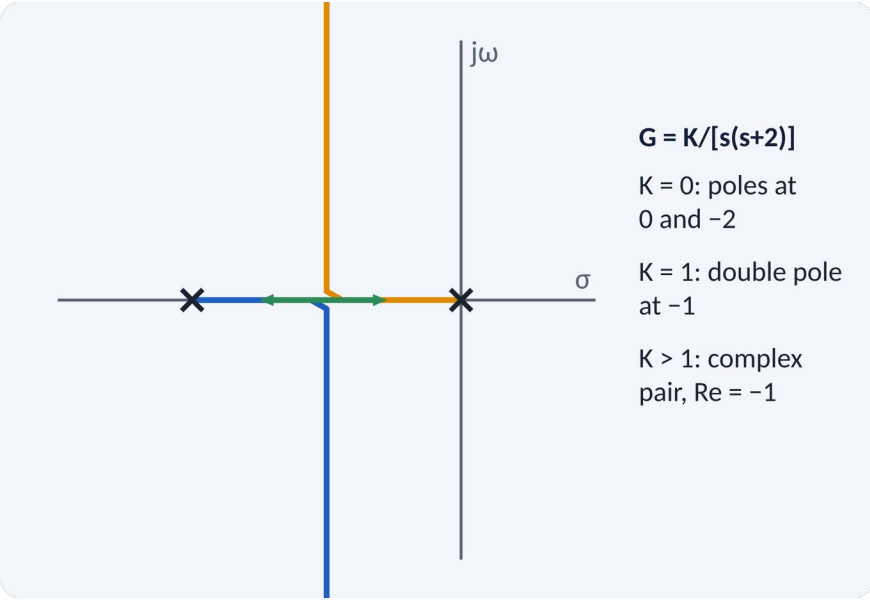
- Closed-loop poles solve 1 + GH = 0.
- Pole position sets the transient response.
- Gain K changes the pole positions.
- Routh gives the stable range of K.



Closed-loop poles: roots of 1 + K·G(s)H(s) = 0, with K ≥ 0

Motivation: Tuning the Gain

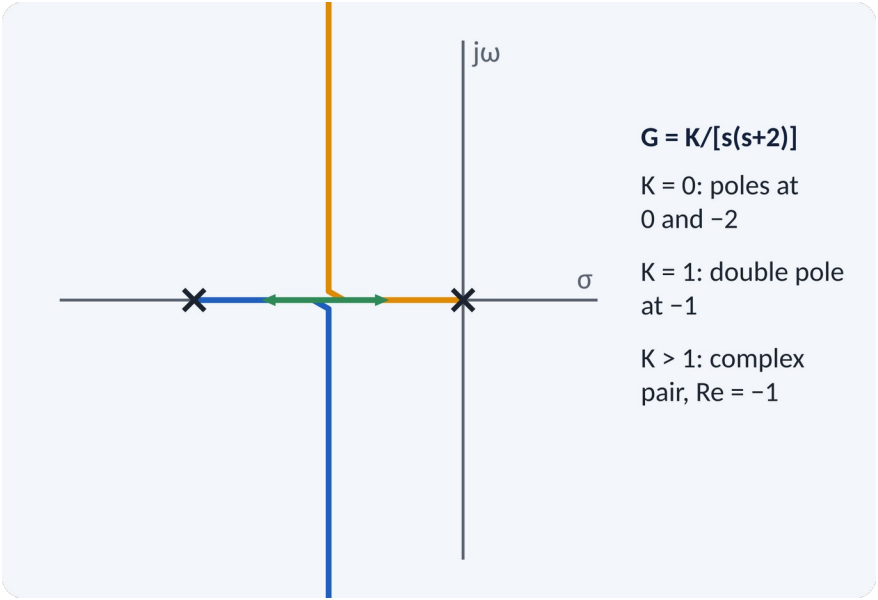
- Gain sets speed, overshoot and stability.
- Trial and error with step tests is slow.
- The locus shows all designs in one picture.
- Choose K to put poles where specs need them.



Root locus = plot of closed-loop poles as K varies from 0 to ∞

Root Locus Concept

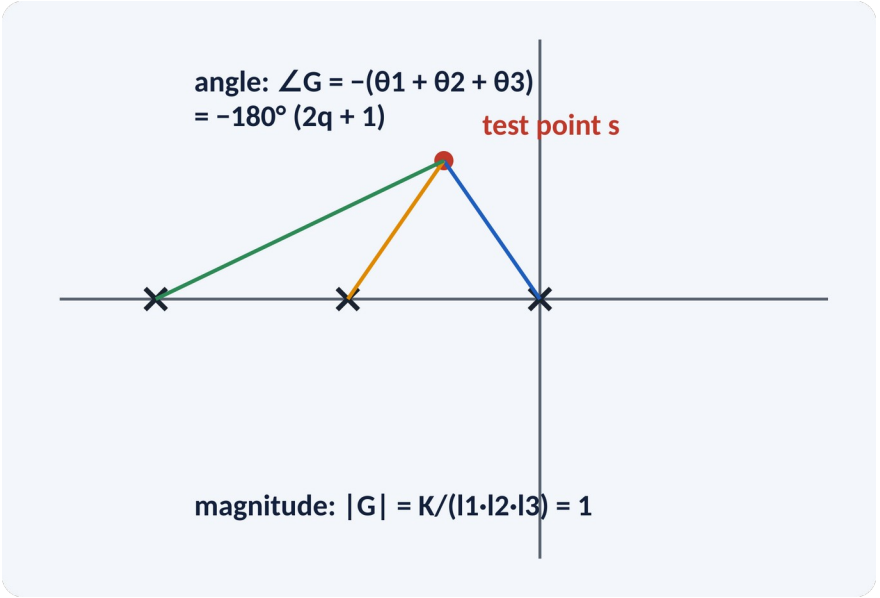
- Poles of G(s)H(s) are the starting points (K = 0).
- Zeros of G(s)H(s) are the end points (K → ∞).
- Each pole traces a branch as K increases.
- A branch leaving to ∞ follows an asymptote.



1 + K·G(s)H(s) = 0 ⇒ K·G(s)H(s) = −1
G(s)H(s) = N(s)/D(s), K gain, K ≥ 0

Magnitude and Angle Conditions

- A point s is on the locus if it satisfies both.
- Angle condition: ∠GH = ±180°(2q + 1).
- Magnitude condition gives the gain K at s.
- Measure with vectors from poles and zeros.



∠G(s)H(s) = ±180°(2q + 1); K = 1/|G(s)H(s)| = (product of pole distances)/(product of zero distances)

Rules 1–2: Branches, Start, End, Symmetry

- Number of branches = number of poles (if P ≥ Z).
- Branches start at poles, end at zeros or ∞.
- P − Z branches go to infinity.
- The locus is symmetrical about the real axis.

Branches
number = max(P, Z) (P poles, Z zeros of GH)

Start (K = 0)
at the open-loop poles

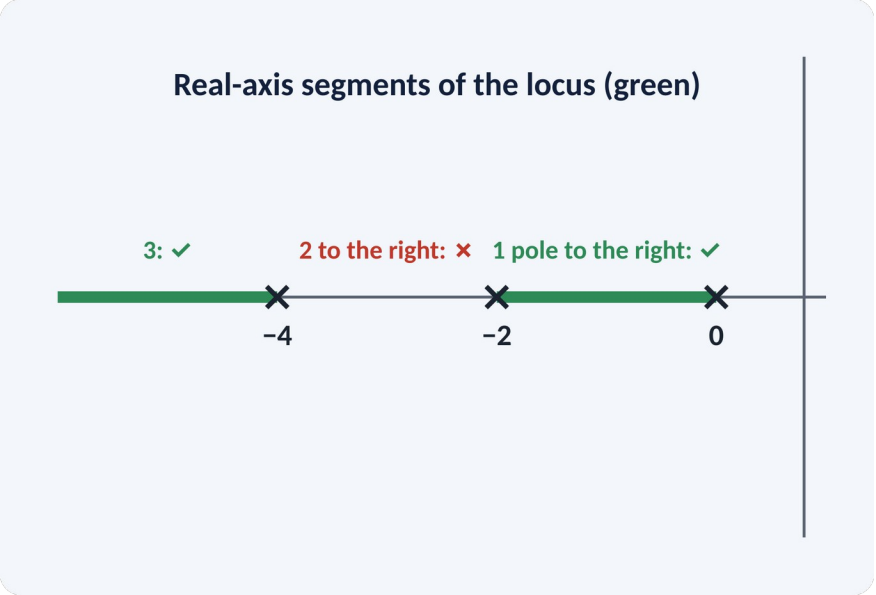
End (K → ∞)
at the open-loop zeros, or at infinity

Symmetry
locus is symmetric about the real axis

P poles, Z zeros → P − Z asymptotic branches

Rule 3: Real-Axis Segments

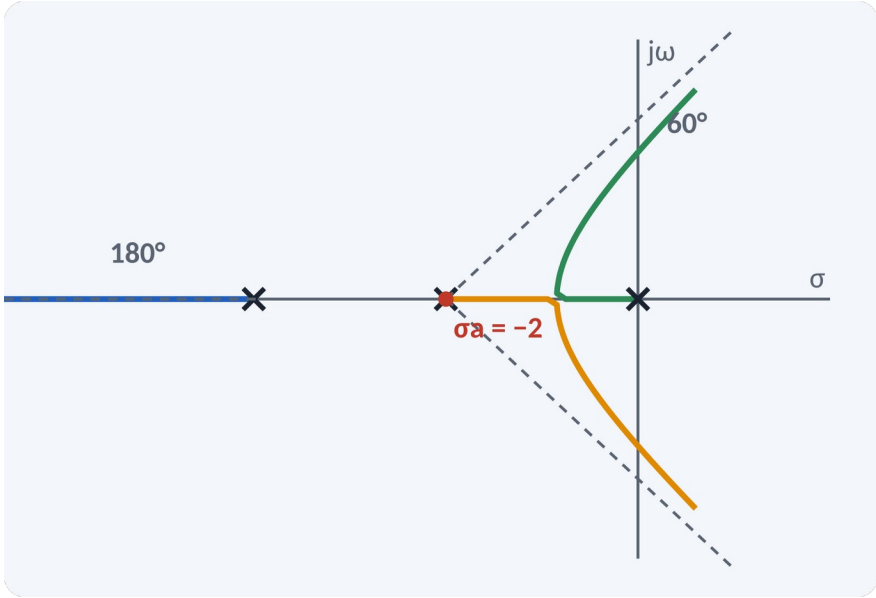
- Take a test point on the real axis.
- Count poles and zeros to its right.
- Odd count: the point is on the locus.
- Even count: not on the locus.



Example poles 0, −2, −4: locus on [−2, 0] and (−∞, −4]

Rule 4: Asymptotes

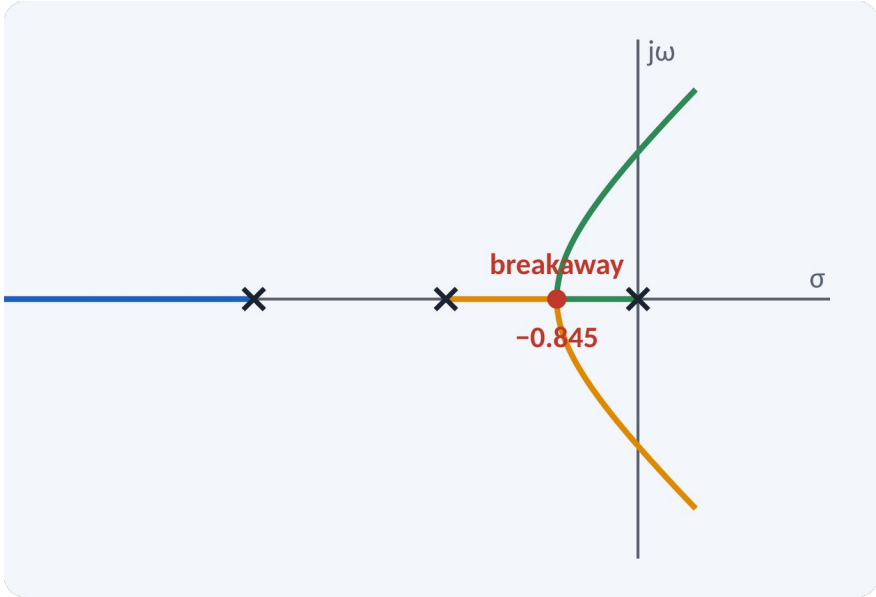
- P − Z branches approach straight lines.
- Angles are equally spaced.
- All asymptotes meet at the centroid σa on the real axis.
- Example P = 3, Z = 0: 60°, 180°, 300°.



θq = ±180°(2q + 1)/(P − Z); σa = (Σ poles − Σ zeros)/(P − Z)
Example: σa = (0 − 2 − 4)/3 = −2

Rule 5: Breakaway and Break-in Points

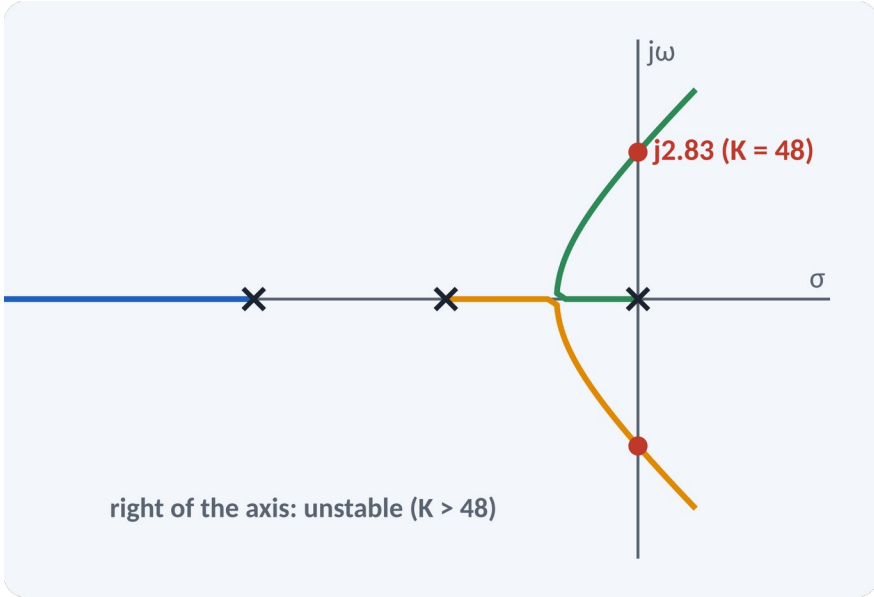
- Branches leave or join the real axis.
- Write K = −1/(G H) as a function of s.
- Solve dK/ds = 0.
- Accept roots lying on the locus (rule 3).
- Compute K at the point.



dK/ds = 0 with K = −D(s)/N(s)
Example: K = −(s³ + 6s² + 8s) → 3s² + 12s + 8 = 0 → s = −0.845 (valid), −3.155 (not on locus)

Rules 6–7: Departure Angle and jω Crossing

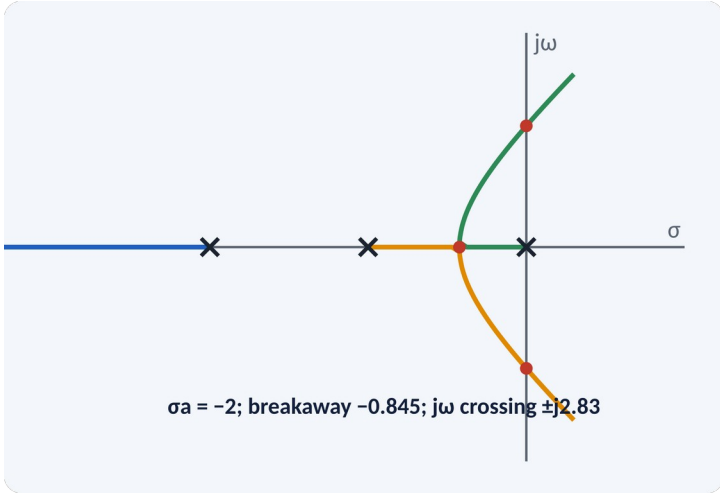
- Departure angle from a complex pole: 180° − Σ(other angles).
- Arrival angle at a complex zero: similar.
- jω crossing: put s = jω or use Routh.
- Crossing gives K for marginal stability.



Departure angle = 180° − Σ ∠(poles) + Σ ∠(zeros)
Crossing: Routh row of zeros → K = 48, ω = 2.83 rad/s

Solved Example 1: Sketch the Locus

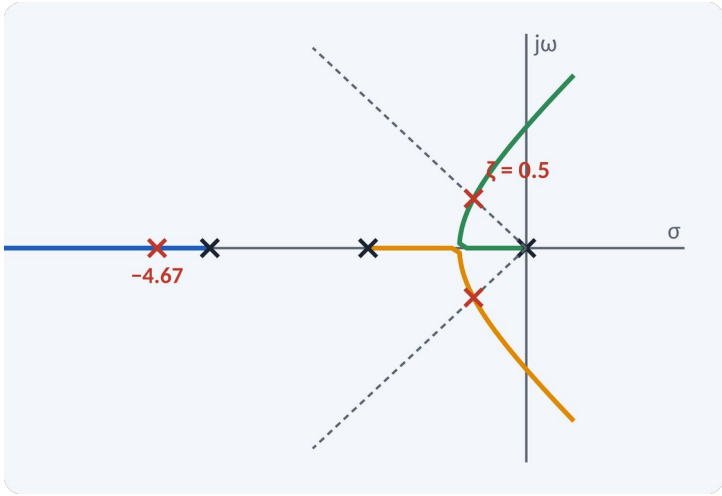
- Sketch the locus of G(s)H(s) = K/[s(s + 2)(s + 4)].
- 1. P = 3, Z = 0: three branches to ∞.
- 2. Real axis: [−2, 0] and (−∞, −4].
- 3. Asymptotes 60°, 180°, 300°; σa = −2.
- 4. Breakaway: $3s^2 + 12s + 8 = 0 \rightarrow s = -0.845$, K = 3.08.
- 5. jω crossing: K = 48, s = ±j2.83.
- 6. Complete the sketch.



Answer: breakaway −0.845 (K = 3.08); unstable for K > 48

Solved Example 2: Gain for ζ = 0.5

- For the same system find K for ζ = 0.5 and the closed-loop poles.
- 1. ζ = 0.5 → poles on the line at 60° from the negative real axis.
- 2. Intersection with the locus: s = −0.667 ± j1.155.
- 3. Third pole: sum of poles = −6 → −4.67.
- 4. K = |s||s + 2||s + 4| = 1.333 × 1.764 × 3.53 ≈ 8.3.



Answer: K ≈ 8.3; poles −0.667 ± j1.155 and −4.67

Common Mistakes and Exam Tips

Common mistakes

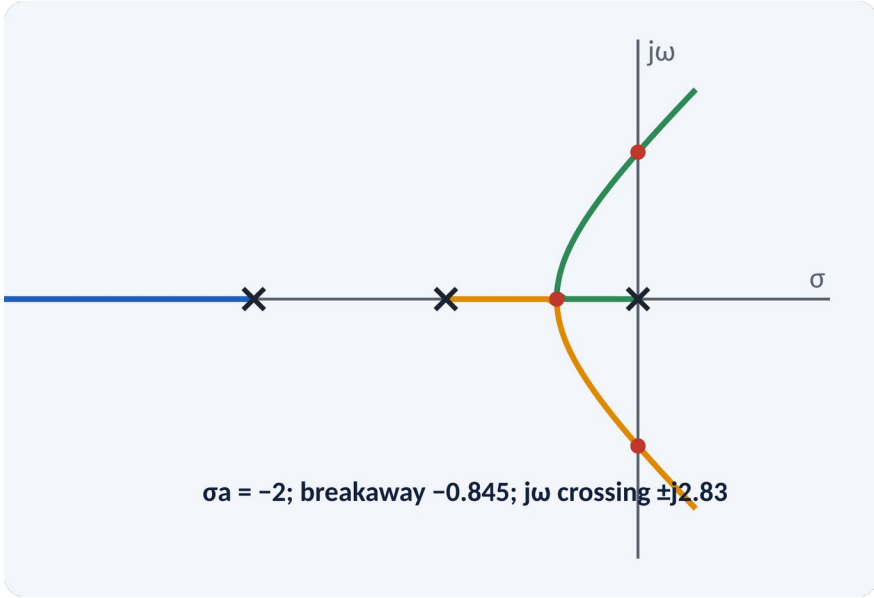
- Counting only poles to the left for real-axis segments.
- Accepting a breakaway root that is not on the locus.
- Mixing up σa formula signs.
- Taking asymptote angles from the positive axis only.
- Drawing branches that cross each other.

Exam tips

- Follow the rules in order.
- Check each candidate point with the angle condition.
- Verify K at the jω crossing with Routh.
- Remember the locus is symmetric about the real axis.

Summary

- Root locus = path of closed-loop poles for 0 ≤ K < ∞.
- Angle condition finds the locus; magnitude gives K.
- Real-axis segments: odd number of poles + zeros to the right.
- Asymptotes: angles 180°(2q + 1)/(P − Z), centroid σa.
- Breakaway: dK/ds = 0; jω crossing from Routh.



Choose K from the locus to meet ζ, ts and stability requirements

Quick Quiz

Q1. Root locus branches start at:
(a) zeros (b) poles of GH (c) origin (d) infinity

Ans: (b)

Q2. Number of asymptotes for P = 4, Z = 1:
(a) 1 (b) 3 (c) 4 (d) 5

Ans: (b)

Q3. A real-axis point is on the locus if poles + zeros to its right is:
(a) even (b) odd (c) zero (d) any

Ans: (b)

Q4. Centroid for poles 0, −2, −4 and no zeros:
(a) −6 (b) −3 (c) −2 (d) 0

Ans: (c)

Q5. Breakaway points satisfy:
(a) $dK/ds = 0$ (b) $K = 0$ (c) $|GH| = 0$ (d) $s = j\omega$

Ans: (a)

References

1. **Benjamin C. Kuo, Automatic Control Systems**
   7th Ed., PHI, 2007 (Units 3, 4, 5 textbook)

2. **M. Gopal, Control Systems Principles and Design**
   4th Ed., TMH, 1999 (reference, Units 4, 5)

3. **K. Ogata, Modern Control Engineering**
   5th Ed., Pearson Education Asia / PHI, 2009 (reference)

Next lecture: Unit 5: compensators and controllers